

Chapter 2: Angles

11 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Measurement and space: MA4-ANG-C-01 applies angle relationships to solve problems, including those related to transversals on sets of parallel lines.

NAME _____

CLASS _____

CHAPTER

2

2.01 Naming angles

2.02 Measuring and drawing angles

2.03 Classifying angles

2.04 Complementary and supplementary angles

2.05 Angles at a point and vertically opposite angles

2.06 Parallel and perpendicular lines

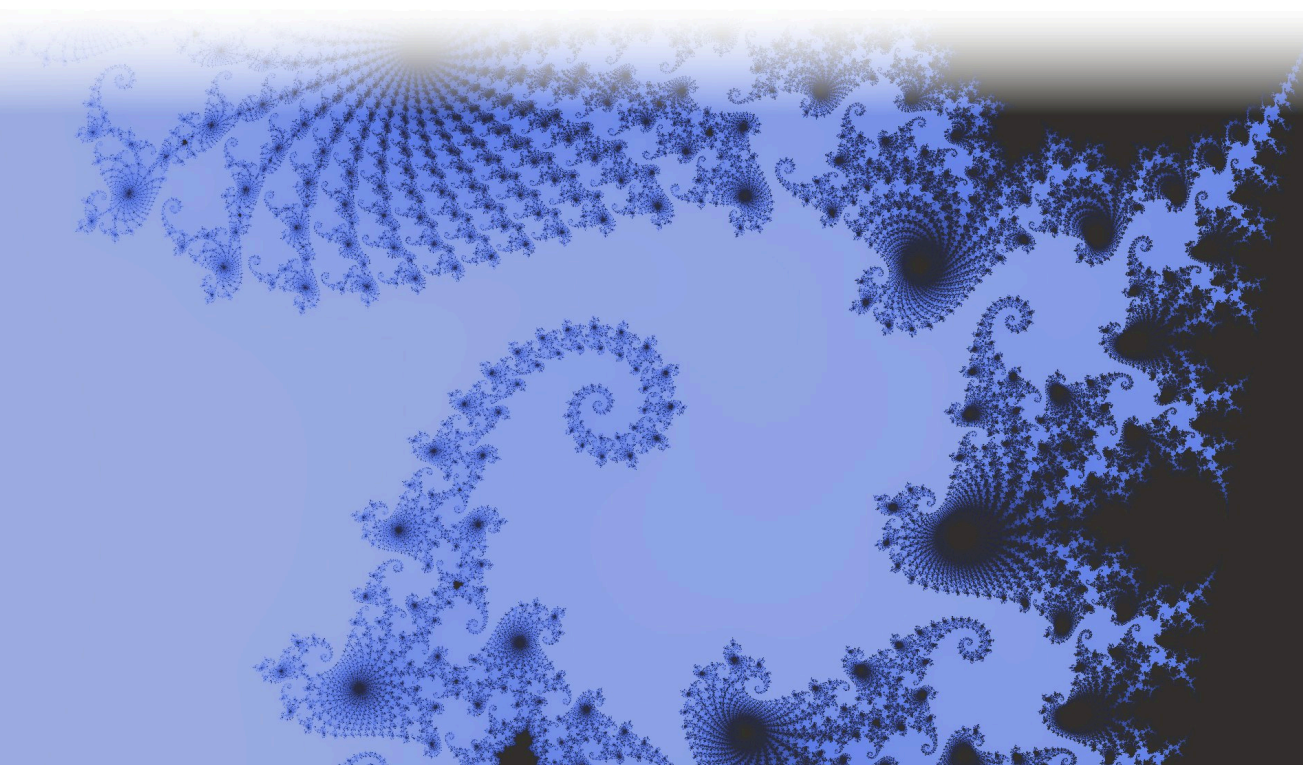
2.07 Corresponding angles

2.08 Alternate angles

2.09 Co-interior angles

2.10 Finding unknown angles on parallel lines

2.11 Proving lines are parallel



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL

Skill drill pages. After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

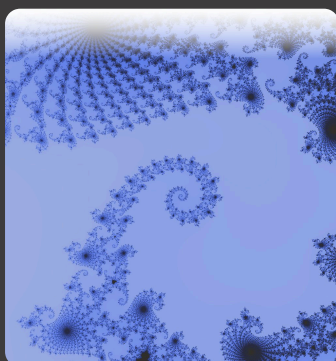
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
2.01	Naming angles	3		☐
2.02	Measuring and drawing angles	9		☐
2.03	Classifying angles	15		☐
2.04	Complementary and supplementary angles	21		☐
2.05	Angles at a point and vertically opposite angles	27		☐
2.06	Parallel and perpendicular lines	33		☐
2.07	Corresponding angles	39		☐
2.08	Alternate angles	45		☐
2.09	Co-interior angles	51		☐
2.10	Finding unknown angles on parallel lines	57		☐
2.11	Proving lines are parallel	63		☐

In this chapter you will

- measure and draw angles with a protractor
- classify angles by size
- find complementary, supplementary and vertically opposite angles
- construct parallel and perpendicular lines
- find corresponding, alternate and co-interior angles on parallel lines
- prove that two lines are parallel



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is Seahorse Valley, the curling tails between the main body and the round bulb on its left. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
2.01

Naming angles

Date _____

Learning intentions

We are learning to:

- name an angle using three letters
- identify adjacent angles

Success criteria

I can:

- ☐ name an angle with the **vertex** in the middle
- ☐ draw an angle from its name
- ☐ name an **adjacent** angle

Key terms Write the definition as your teacher explains it.

Vertex

Arm

Adjacent angles

Notes

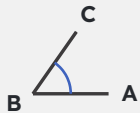
Level 1 · Getting started

Easy · 21 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Name the angle, then draw an angle from its name.

a Name this angle in two ways.



b Draw and label $\angle BKT$.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 21 QUESTIONS

REMEMBER An angle has two **ARMS** and a **VERTEX**.

ROUND 1 Questions 1–12: which letter is the vertex of each angle?

1 $\angle SXF$

2 $\angle UDF$

3 $\angle LDZ$

4 $\angle UBA$

5 $\angle ZYB$

6 $\angle TNU$

7 $\angle TER$

8 $\angle EWH$

9 $\angle PDQ$

10 $\angle SVD$

11 $\angle KSM$

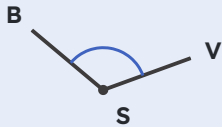
12 $\angle PAC$

YOU DO On your own: **Easy** keep going: Round 2.

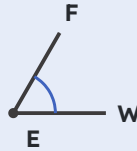
9 QUESTIONS MORE

ROUND 2 Questions 13–21: name the marked angle with three letters (vertex in the middle).

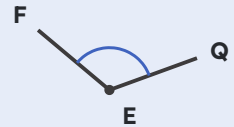
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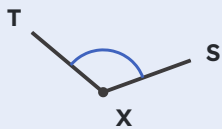
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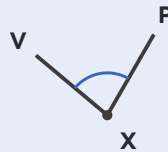
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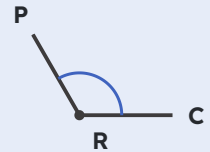
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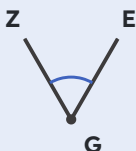
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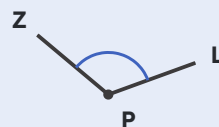
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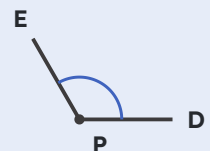
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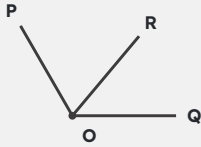
Level 2 · Building up

Medium · 20 questions

WE DO With your teacher: copy what they write on the board.

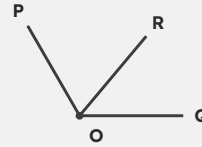
Example 2 Name the angles at O.

a Name the angle adjacent to $\angle POQ$.



Your teacher works this out. Copy every step.

b Name all 3 angles at O.



Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER Name it with 3 letters; the vertex goes in the middle.

ROUND 1 Questions 1–20: write the two arms of each angle.

1 $\angle MZP$

2 $\angle MCS$

3 $\angle UNX$

4 $\angle YQH$

5 $\angle VPZ$

6 $\angle ELM$

7 $\angle ENU$

8 $\angle SAK$

9 $\angle CQN$

10 $\angle VPD$

11 $\angle LUY$

12 $\angle EYG$

13 $\angle LWE$

14 $\angle ZVM$

15 $\angle QRY$

16 $\angle XWS$

17 $\angle STK$

18 $\angle ZST$

19 $\angle UWL$

20 $\angle YFQ$

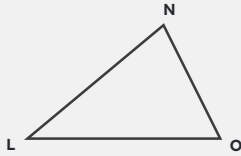
Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Name the marked angle, then answer the question.

a Name the angle marked with a •.



The • is at O.

Your teacher works this out. Copy every step.

b Why is it wrong to call an angle “ $\angle B$ ” when three lines meet at B?

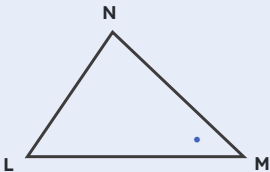
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Name the marked angle in each diagram.

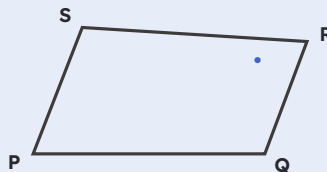
2 QUESTIONS

REMEMBER $\angle ABC$ and $\angle CBA$ are the same angle.

1 The angle at vertex M in $\triangle LMN$.



2 The angle at R in quadrilateral PQRS.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Four rays come from point O. How many different angles are there?

- Count angles using each pair of rays
- With 4 rays there are $3 + 2 + 1$ pairs

Answer: 6 angles

YOU DO

On your own: read the worked example, then try these.

- E1** How many angles are made by 5 rays from one point?

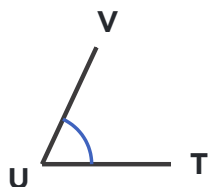
- E2** Name all the angles made by rays OA, OB, OC and OD.

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

Name the angle shown.

Level 2

Which letter is the vertex? Why?

Level 3

Draw and label two adjacent angles $\angle ABC$ and $\angle CBD$.

Lesson summary

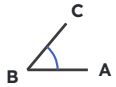
Missed the lesson? Start here.

- Remember**
1. An angle has two **arms** and a **vertex**.
 2. Name it with 3 letters; the vertex goes in the middle.
 3. $\angle ABC$ and $\angle CBA$ are the same angle.
 4. **Adjacent** angles share a vertex and an arm.

Level 1

Name the angle

1. Vertex is B, arms go to A and C



Answer: $\angle ABC$

Level 2

Draw $\angle KLM$

1. Vertex L
2. Arms LK and LM

Answer: a V shape with L at the point

Level 3

Adjacent to $\angle POQ$?

1. Shares arm OQ and vertex O

Answer: $\angle QOR$

Extension

Angles from 4 rays

1. $3 + 2 + 1$

Answer: 6

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.02

Measuring and drawing angles

Date _____

Learning intentions

We are learning to:

- use a protractor to measure and draw angles, including reflex angles

Success criteria

I can:

- ☐ place the protractor centre on the vertex and 0° on one arm
- ☐ read the correct scale
- ☐ draw a **reflex** angle by drawing 360° minus the angle

Key terms Write the definition as your teacher explains it.

Protractor

Reflex angle

Notes

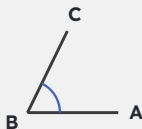
Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Measure the angle, then draw an angle of the given size.

a Measure $\angle ABC$.



b Draw $\angle JUG = 84^\circ$.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 20 QUESTIONS

REMEMBER Centre of the protractor on the vertex.

ROUND 1 Questions 1–8: reflex angles: write the angle you draw with the protractor (360° – the angle).

1 205°

2 255°

3 245°

4 265°

5 290°

6 325°

7 215°

8 190°

YOU DO

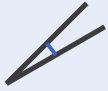
On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 9–20: measure each angle with your protractor.

9



10



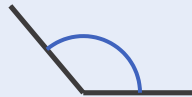
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Level 2 · Building up

Medium · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Measure the angle, then draw an angle of the given size.

a Measure this obtuse angle.



b Draw $\angle QRA = 117^\circ$.



Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

16 QUESTIONS

REMEMBER 0° line along one arm.

ROUND 1 Questions 1–16: draw each angle in your book, then label its type here.

1 210°

2 150°

3 120°

4 275°

5 90°

6 145°

7 320°

8 295°

9 200°

10 25°

11 130°

12 290°

13 110°

14 255°

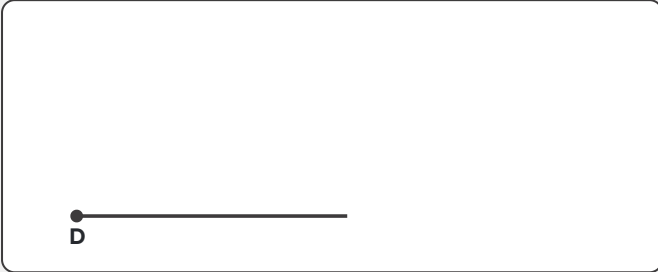
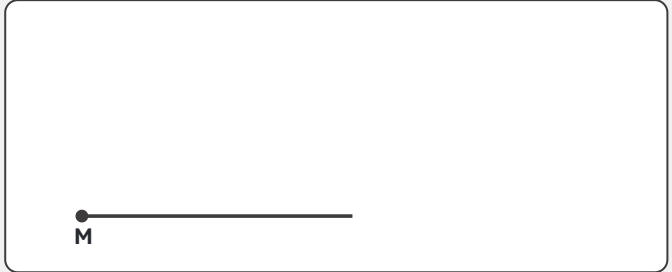
15 5°

16 270°

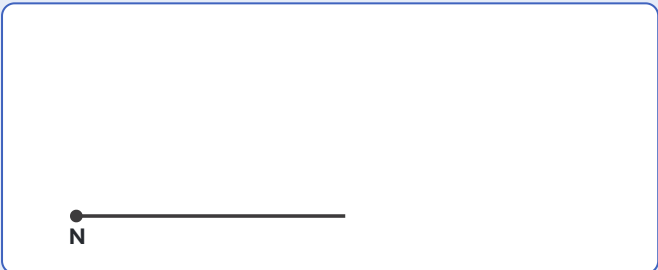
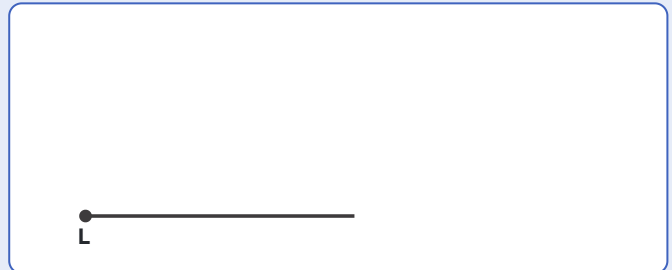


Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.**Example 3** Draw each reflex angle.**a** Draw $\angle XDW = 210^\circ$.**b** Draw $\angle AMP = 300^\circ$.**YOU DO** On your own: **Challenging** Reflex angles.

2 QUESTIONS

REMEMBER Count up from 0° on that scale.**1** Draw $\angle MNB = 195^\circ$ **2** Draw $\angle PLO = 270^\circ$ 

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

The hands of a clock show 4 o'clock. What is the angle between them? What is the reflex angle?

1. Each hour mark is $360 \div 12 = 30^\circ$
2. 4 hours apart: $4 \times 30 = 120^\circ$
3. Reflex: $360 - 120 = 240^\circ$

Answer: 120° and reflex 240°

YOU DO

On your own: read the worked example, then try these.

- E1** Find the angle between the hands at 5 o'clock and at 9 o'clock.

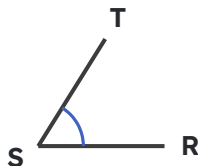
- E2** At 3:30 the hour hand is halfway between 3 and 4. Find the angle between the hands.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

Measure $\angle RST$.

Level 2

Draw $\angle ABC = 135^\circ$.

Level 3

Draw $\angle XYZ = 250^\circ$.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Centre of the protractor on the vertex.
 2. 0° line along one arm.
 3. Count up from 0° on that scale.
 4. Reflex: measure the small angle, then $360^\circ - \text{it}$.

Level 1

Measure a 64° angle

1. Centre on vertex, 0° on arm
2. Read the scale that starts at 0°

Answer: 64°

Level 2

Draw 117°

1. Draw an arm, mark 117° , join to the vertex

Answer: an obtuse angle

Level 3

Draw 210°

1. $360 - 210 = 150$
2. Draw 150° the other way

Answer: a reflex angle

Extension

Angle at 4 o'clock

1. $4 \times 30^\circ$

Answer: 120°

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.03

Classifying angles

Date _____

Learning intentions

We are learning to:

- classify angles as acute, right, obtuse, straight, reflex or a revolution

Success criteria

I can:

- ☐ know the size range for each type of angle
- ☐ classify an angle from its size or a diagram
- ☐ name and classify an angle in a diagram

Key terms Write the definition as your teacher explains it.

Acute

Obtuse

Reflex

Notes

Level 1 · Getting started

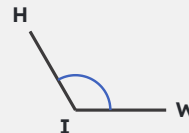
Easy · 21 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Classify each angle.

a Classify: 34° , 90° , 145° , 180° , 250° , 360°

b Name and classify the angle.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 21 QUESTIONS

REMEMBER **ACUTE**: less than 90° . **RIGHT**: 90° .

ROUND 1 Questions 1–12: classify each angle: acute, right, obtuse, straight, reflex or revolution.

1 345°

2 120°

3 260°

4 350°

5 210°

6 90°

7 220°

8 25°

9 335°

10 295°

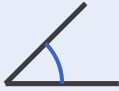
11 180°

12 110°

YOU DO On your own: **Easy** keep going: Round 2.

9 QUESTIONS MORE

ROUND 2 Questions 13–21: classify the angle.



13



14



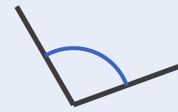
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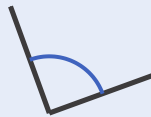
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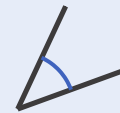
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21

Level 2 · Building up

Medium · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Answer each question.

a Is 90.5° acute, right or obtuse?

Your teacher works this out. Copy every step.

b An angle is 3 times a right angle. Classify it.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER **OBTUSE**: between 90° and 180° .

ROUND 1 Questions 1–16: add the two angles, then classify the total.

1 $25^\circ + 200^\circ$

2 $145^\circ + 160^\circ$

3 $40^\circ + 60^\circ$

4 $90^\circ + 85^\circ$

5 $95^\circ + 165^\circ$

6 $145^\circ + 70^\circ$

7 $20^\circ + 85^\circ$

8 $145^\circ + 150^\circ$

9 $60^\circ + 100^\circ$

10 $110^\circ + 110^\circ$

11 $45^\circ + 115^\circ$

12 $125^\circ + 20^\circ$

13 $195^\circ + 10^\circ$

14 $95^\circ + 130^\circ$

15 $135^\circ + 150^\circ$

16 $145^\circ + 195^\circ$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 17–24: true or false?

17 A right angle is exactly 90° .

18 Two acute angles always make an obtuse angle.

19 An obtuse angle and a right angle can add to 200° .

20 A reflex angle is between 180° and 360° .

21 Half a revolution is a straight angle.

22 359° is reflex.

23 Every obtuse angle is less than 180° .

24 180° is a reflex angle.

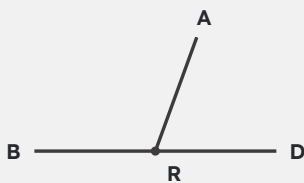
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Name and classify, then explain.

a Name and classify both angles at R.



Your teacher works this out. Copy every step.

b Can a triangle have an obtuse angle and a right angle? Explain.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER **STRAIGHT**: 180° . **REFLEX**: between 180° and 360° .

1 What angle does a minute hand turn through in 20 minutes? Classify it.

2 Two acute angles are added. Can the answer be obtuse? Give an example.

3 List 3 obtuse angles that are multiples of 25° .

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

A wheel turns $1\frac{3}{4}$ revolutions. How many degrees is that?

1. 1 revolution = 360°
2. $\frac{3}{4}$ of $360^\circ = 270^\circ$
3. $360 + 270$

Answer: 630°

YOU DO

On your own: read the worked example, then try these.

- E1** A skater spins $2\frac{1}{2}$ times. How many degrees?

- E2** A fan turns 1080° . How many revolutions?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Classify 88° .

Level 2

Classify 200° and 180° .

Level 3

Name and classify both angles in Example 5 ($\angle DRA$ and $\angle ARB$).

Lesson summary

Missed the lesson? Start here.

- Angle types**
1. **Acute:** less than 90° . **Right:** 90° .
 2. **Obtuse:** between 90° and 180° .
 3. **Straight:** 180° . **Reflex:** between 180° and 360° .
 4. **Revolution:** 360° .

Level 1

Classify 125°

1. Between 90° and 180°

Answer: obtuse

Level 2

Classify 300°

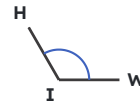
1. Between 180° and 360°

Answer: reflex

Level 3

Name and classify

1. Vertex I in the middle
2. More than 90°



Answer: $\angle WIH$, obtuse

Extension

$1\frac{3}{4}$ revolutions

1. $360 + 270$

Answer: 630°

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.04

Complementary and supplementary angles

Date _____

Learning intentions

We are learning to:

- find complementary and supplementary angles

Success criteria

I can:

- ☐ know complementary angles add to 90°
- ☐ know supplementary angles add to 180°
- ☐ find an unknown angle in a right angle or on a straight line

Key terms Write the definition as your teacher explains it.

Complementary

Supplementary

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the complement and the supplement.

a Find the complement of 35° and of 78° .

b Find the supplement of 45° and of 100° .

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER **COMPLEMENTARY** angles add to 90° .

ROUND 1 Questions 1–12: find the complement (they add to 90°).

1 14°

2 34°

3 37°

4 36°

5 66°

6 49°

7 6°

8 54°

9 23°

10 20°

11 80°

12 1°

YOU DO

On your own: **Easy** keep going: Round 2.

20 QUESTIONS MORE

ROUND 2

Questions 13–32: find the supplement (they add to 180°).

13 107°

14 48°

15 77°

16 18°

17 1°

18 125°

19 74°

20 178°

21 54°

22 166°

23 159°

24 161°

25 165°

26 53°

27 39°

28 13°

29 81°

30 62°

31 113°

32 90°

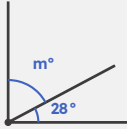
Level 2 · Building up

Medium · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the value of the pronumeral.

a Find the value of m .



Your teacher works this out. Copy every step.

b Find the value of k .



Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER SUPPLEMENTARY angles add to 180° .

ROUND 1 Questions 1–20: are the two angles complementary (C), supplementary (S) or neither (N)?

1 43° and 47° _____

2 153° and 169° _____

3 57° and 33° _____

4 86° and 97° _____

5 86° and 4° _____

6 79° and 11° _____

7 91° and 89° _____

8 88° and 2° _____

9 161° and 19° _____

10 72° and 108° _____

11 39° and 51° _____

12 170° and 10° _____

13 152° and 140° _____

14 151° and 29° _____

15 116° and 64° _____

16 49° and 119° _____

17 49° and 41° _____

18 147° and 33° _____

19 119° and 9° _____

20 37° and 42° _____

YOU DO On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2 Questions 21–32: find the value of the pronumeral.

- | | | |
|---|--|---|
| 21 86° and x° are supplementary
_____ | 22 152° and x° are supplementary
_____ | 23 67° and k° are complementary
_____ |
| 24 77° and m° are complementary
_____ | 25 40° and x° are complementary
_____ | 26 26° and m° are complementary
_____ |
| 27 44° and p° are complementary
_____ | 28 105° and k° are supplementary
_____ | 29 93° and x° are supplementary
_____ |
| 30 63° and x° are supplementary
_____ | 31 37° and y° are complementary
_____ | 32 73° and k° are supplementary
_____ |

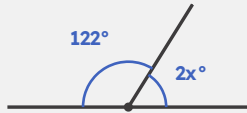
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find x , then answer the question.

a Find x .



Your teacher works this out. Copy every step.

b Two supplementary angles are equal. What size are they?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Harder problems.

3 QUESTIONS

REMEMBER Right angle split in two → complementary.

- | | | |
|--|--|--|
| 1 An angle is 20° more than its complement. Find both angles.
_____ | 2 The supplement of an angle is 4 times the angle. Find the angle.
_____ | 3 Find x if $3x^\circ$ and 60° are supplementary.
_____ |
|--|--|--|



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Find the angle whose supplement is 3 times its complement.

1. Let the angle be x
2. $180 - x = 3(90 - x)$
3. $180 - x = 270 - 3x$, so $2x = 90$

Answer: $x = 45^\circ$

YOU DO

On your own: read the worked example, then try these.

- E1 Find the angle whose supplement is 4 times its complement.

- E2 Find the complement of the supplement of 130° .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Find the complement of 62° .

Level 2

Find the supplement of 118° .

Level 3

Two angles on a straight line are x° and $3x^\circ$. Find x .

Lesson summary

Missed the lesson? Start here.

Remember

1. **Complementary** angles add to 90° .
2. **Supplementary** angles add to 180° .
3. Right angle split in two → complementary.
4. Straight line split in two → supplementary.

Level 1

Complement of 35°

1. $90 - 35$

Answer: 55°

Level 2

Supplement of 100°

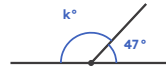
1. $180 - 100$

Answer: 80°

Level 3

Find k

1. $k + 47 = 180$



Answer: $k = 133$

Extension

Supplement = $3 \times$ complement

1. $180 - x = 270 - 3x$

Answer: 45°

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
2.05

Angles at a point and vertically opposite angles

Date _____

Learning intentions

We are learning to:

- find unknown angles using angle rules

Success criteria

I can:

- ☐ use angles on a straight line add to 180°
- ☐ use angles at a point add to 360°
- ☐ use vertically opposite angles are equal

Key terms Write the definition as your teacher explains it.

Angles at a point

Vertically opposite

Notes

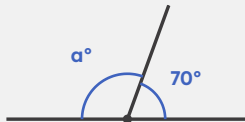
Level 1 · Getting started

Easy · 28 questions

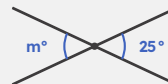
WE DO With your teacher: copy what they write on the board.

Example 1 Find the value of each pronumeral. Give a reason.

a Find a .



b Find m .



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

REMEMBER Straight line: angles add to 180° .

ROUND 1 Questions 1–12: angles on a straight line add to 180° . Find a .

1 110° and a°

2 81° and a°

3 30° and a°

4 78° and a°

5 21° and a°

6 16° and a°

7 18° and a°

8 26° and a°

9 141° and a°

10 64° and a°

11 96° and a°

12 52° and a°

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 13–28: angles at a point add to 360° . Find a .

13 $125^\circ, 132^\circ$ and a°

14 $135^\circ, 93^\circ$ and a°

15 $55^\circ, 103^\circ$ and a°

16 $127^\circ, 148^\circ$ and a°

17 $80^\circ, 63^\circ$ and a°

18 $164^\circ, 168^\circ$ and a°

19 $53^\circ, 133^\circ$ and a°

20 $165^\circ, 169^\circ$ and a°

21 $54^\circ, 118^\circ$ and a°

22 $113^\circ, 45^\circ$ and a°

23 $150^\circ, 148^\circ$ and a°

24 $50^\circ, 89^\circ$ and a°

25 $126^\circ, 111^\circ$ and a°

26 $83^\circ, 122^\circ$ and a°

27 $57^\circ, 124^\circ$ and a°

28 $127^\circ, 64^\circ$ and a°

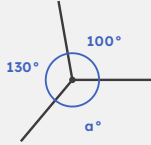
Level 2 · Building up

Medium · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Angles at a point: find the pronumeral.

a Find a (angles at a point).



b Find x .



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER At a point: angles add to 360° .

ROUND 1

Questions 1–12: two lines cross. One angle is given. Find the vertically opposite angle (v) and the angle next to it (n).

1 77°

2 84°

3 76°

4 127°

5 130°

6 83°

7 34°

8 74°

9 70°

10 61°

11 91°

12 136°

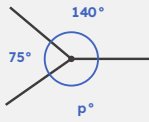
Level 3 · Challenge

Challenging · 2 questions · then try the extension

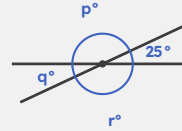
WE DO With your teacher: copy what they write on the board.

Example 3 Find each pronumeral. Give a reason.

a Find p .



b Find p , q and r .



Your teacher works this out. Copy every step.

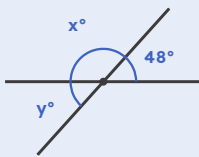
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Find each variable. Give a reason.

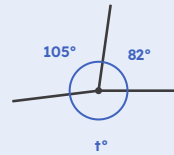
2 QUESTIONS

REMEMBER VERTICALLY OPPOSITE angles are equal.

1 Find x and y .



2 Find t .



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Angles at a point are $2x^\circ$, $3x^\circ$ and $4x^\circ$. Find x .

1. Angles at a point add to 360°
2. $2x + 3x + 4x = 9x = 360$

Answer: $x = 40$

YOU DO

On your own: read the worked example, then try these.

- E1** Angles on a straight line are x° , $2x^\circ$ and $3x^\circ$. Find each angle.

- E2** Two lines cross. One angle is $(2x + 10)^\circ$ and its vertically opposite angle is 70° . Find x .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Angles on a straight line: 64° and x° . Find x .

Level 2

Two lines cross. One angle is 38° . Find the vertically opposite angle.

Level 3

Angles at a point: 95° , 130° and y° . Find y .

Lesson summary

Missed the lesson? Start here.

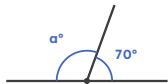
Remember

1. Straight line: angles add to **180°**.
2. At a point: angles add to **360°**.
3. **Vertically opposite** angles are equal.
4. Always give a reason.

Level 1

Find a

1. $a + 70 = 180$

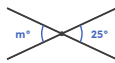


Answer: $a = 110$

Level 2

Find m

1. Vertically opposite



Answer: $m = 25$

Level 3

At a point: $140^\circ, 75^\circ, p^\circ$

1. $p = 360 - 215$

Answer: $p = 145$

Extension

$2x + 3x + 4x = 360$

1. $9x = 360$

Answer: $x = 40$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON 2.06

Parallel and perpendicular lines

Date _____

Learning intentions

We are learning to:

- identify parallel and perpendicular lines
- construct parallel and perpendicular lines with compasses

Success criteria

I can:

- ☐ use the symbols $\parallel$ and $\perp$
- ☐ construct a line through a point parallel to another line
- ☐ construct a line through a point perpendicular to another line

Key terms Write the definition as your teacher explains it.

Parallel

Perpendicular

Intersect

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Name the lines, then explain the symbols.

- a** Name 2 parallel lines and 2 perpendicular lines. **b** What does $AB \perp CD$ mean? What does $EF \parallel GH$ mean?



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

18 QUESTIONS

REMEMBER PARALLEL ($\parallel$): same direction, never meet.

ROUND 1 Questions 1–6: write the symbol for each statement: $\parallel$ (parallel) or $\perp$ (perpendicular).

- 1

PQ is perpendicular to RS

2

the hands of a clock at 3:00

3

TU is at right angles to VW

4

the top and a side of a door

5

XY is parallel to ZW

6

KL never meets MN

YOU DO

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 7–18: parallel (P), perpendicular ($\perp$) or neither (N)?

7 a letter T

8 a plus sign

9 an equals sign

10 a letter Z (top and bottom)

11 a letter L

12 the edges of a book (top and side)

13 a letter H (the two sides)

14 the lanes of a pool

15 the sides of a ladder

16 a letter K

17 a letter X

18 a letter V

Level 2 · Building up

Medium · 12 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Construct the line. Use a ruler and compasses.**a** Construct a line through P parallel to BC.**b** Construct a line through X perpendicular to BC (X on BC).**YOU DO**On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER PERPENDICULAR ($\perp$): meet at 90° .**ROUND 1** Questions 1–12: true or false?**1** A square has two pairs of parallel sides.

2 Arrows on lines show they are parallel.

3 Parallel lines are the same distance apart everywhere.

4 Perpendicular lines meet at 45° .

5 Vertical and horizontal lines are perpendicular.

6 The symbol $\perp$ means perpendicular.

7 A rectangle has 4 pairs of perpendicular sides.

8 Parallel lines can cross.

9 Lines that meet at 89° are perpendicular.

10 Two lines perpendicular to the same line are parallel.

11 A transversal crosses two or more lines.

12 A triangle can have two parallel sides.



Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Construct the line, then give examples.

- a** Construct a line through X perpendicular to BC (X not on BC).

- b** Name everyday examples of parallel and perpendicular lines.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Constructions.

2 QUESTIONS

REMEMBER Lines that cross **INTERSECT**.

- 1** Draw a line BC and point X below it. Construct the perpendicular from X to BC.

- 2** Construct a square with side 4 cm using perpendicular lines.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Construct the perpendicular bisector of a 6 cm interval AB.

1. Set compasses to more than half of AB
2. Draw arcs above and below AB from A and from B
3. Join the two crossing points

Answer: a line perpendicular to AB through its midpoint

YOU DO

On your own: read the worked example, then try these.

- E1** Draw a 7 cm interval and construct its perpendicular bisector.

- E2** Why does the construction work? (Hint: equal distances.)

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What symbol means “is parallel to”?

Level 2

Name 2 perpendicular lines in your classroom.

Level 3

Construct a line through a point P perpendicular to a line (draw your own).

Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Parallel** ($\parallel$): same direction, never meet.
 2. **Perpendicular** ($\perp$): meet at 90° .
 3. Lines that cross **intersect**.
 4. Constructions use compass arcs of equal radius.

Level 1

Rectangle ABCD

1. Opposite sides

Answer: $AB \parallel DC$, $AB \perp BC$

Level 2

Parallel through P

1. Copy the angle or use two equal arcs

Answer: a line through P that never meets BC

Level 3

Perpendicular through X on BC

1. Equal arcs either side of X
2. Arcs above from those points

Answer: join X to the crossing point

Extension

Perpendicular bisector

1. Arcs from both ends

Answer: join the crossings

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.07

Corresponding angles

Date _____

Learning intentions

We are learning to:

- identify corresponding angles on parallel lines

Success criteria

I can:

- ☐ find **corresponding** angles (F shape)
- ☐ know they are equal on parallel lines
- ☐ find unknown angles using them

Key terms Write the definition as your teacher explains it.

Transversal

Corresponding angles

Notes

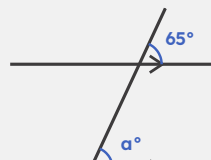
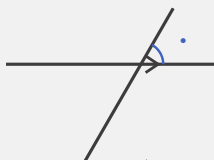
Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Mark the corresponding angle, then find the pronumeral.

- a** Mark the angle corresponding to the one shown. **b** Find a .



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER A **TRANSVERSAL** crosses two or more lines.

ROUND 1 Questions 1–12: the lines are parallel. Corresponding angles are equal. Find the corresponding angle.

1 89°

2 104°

3 23°

4 136°

5 93°

6 55°

7 100°

8 44°

9 28°

10 143°

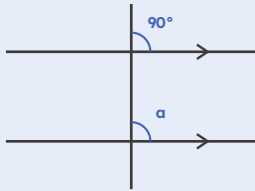
11 64°

12 121°

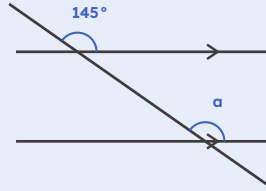
YOU DO On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

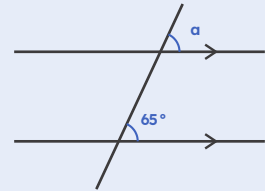
ROUND 2 Questions 13–24: the lines are parallel. Find a .



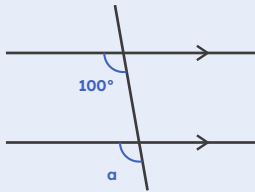
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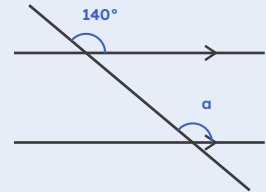
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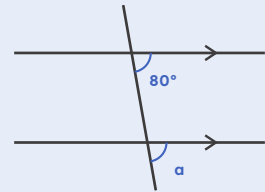
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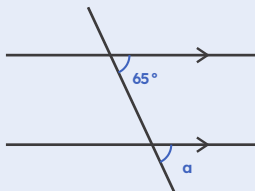
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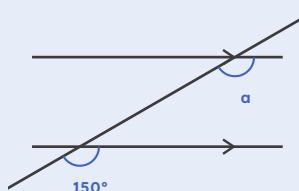
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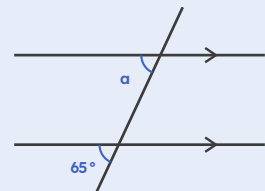
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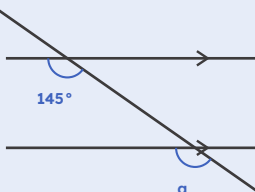
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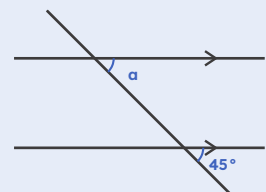
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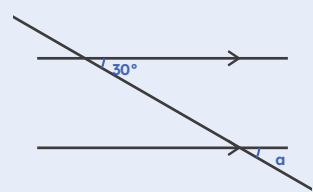
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23



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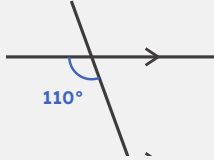
Level 2 · Building up

Medium · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the pronumeral, then answer the question.

a Find m .



b Where is the F shape in corresponding angles?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

REMEMBER **CORRESPONDING** angles are in matching positions (F shape).

ROUND 1 Questions 1–12: the lines are parallel. Find a , then the angle next to it on the line (b).

1 corresponding to 70°

2 corresponding to 85°

3 corresponding to 90°

4 corresponding to 150°

5 corresponding to 35°

6 corresponding to 135°

7 corresponding to 145°

8 corresponding to 60°

9 corresponding to 105°

10 corresponding to 30°

11 corresponding to 125°

12 corresponding to 40°

YOU DO On your own: **Medium** keep going: the next rounds.

16 QUESTIONS MORE

ROUND 2 Questions 13–20: corresponding angles are equal. Find x .

13 $4x + 14 = 142$

14 $2x + 4 = 32$

15 $4x + 27 = 155$

16 $3x + 7 = 55$

17 $3x + 29 = 56$

18 $3x + 30 = 111$

19 $3x + 30 = 63$

20 $3x + 2 = 50$

ROUND 3 Questions 21–28: are these corresponding angles on parallel lines? Write yes or no.

21 82° and 98°

22 78° and 102°

23 38° and 38°

24 100° and 100°

25 131° and 49°

26 105° and 75°

27 105° and 105°

28 54° and 126°

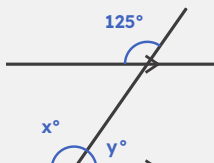
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find x and y , then answer the question.

a Find x and y .



Your teacher works this out. Copy every step.

b Are corresponding angles always equal? Explain.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER On parallel lines they are **EQUAL**.

1 Explain what a transversal is.

2 Draw two parallel lines and a transversal. Mark a pair of corresponding angles.

3 If the lines are not parallel, are corresponding angles equal?

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Corresponding angles are $(3x + 5)^\circ$ and 50° .
Find x .

1. Corresponding angles on parallel lines are equal
2. $3x + 5 = 50$
3. $3x = 45$

Answer: $x = 15$

YOU DO

On your own: read the worked example, then try these.

- E1** Corresponding angles are $(2y - 10)^\circ$ and 70° .
Find y .

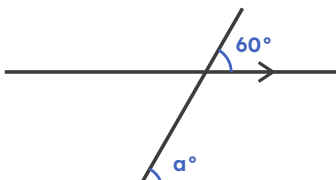
- E2** Corresponding angles are $4a^\circ$ and $(a + 90)^\circ$.
Find a .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

Find a .

Level 2

What shape do corresponding angles make?

Level 3

Corresponding angles are $(2x + 20)^\circ$ and 60° .
Find x .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **transversal** crosses two or more lines.
 2. **Corresponding** angles are in matching positions (F shape).
 3. On parallel lines they are **equal**.
 4. Reason: “corresponding angles, $AB \parallel CD$ ”.

Level 1

Corresponding to top-right?

1. Same position at the other line

Answer: bottom-right

Level 2

Find a

1. Corresponding angles



Answer: $a = 65$

Level 3

Find x then y

1. $x = 125$ corresponding
2. $y = 180 - 125$

Answer: $x = 125, y = 55$

Extension

$$3x + 5 = 50$$

1. $3x = 45$

Answer: $x = 15$

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
2.08

Alternate angles

Date _____

Learning intentions

We are learning to:

- identify alternate angles on parallel lines

Success criteria

I can:

- ☐ find **alternate** angles (Z shape)
- ☐ know they are equal on parallel lines
- ☐ find unknown angles using them

Key terms Write the definition as your teacher explains it.

Alternate angles

Z shape

Notes

Level 1 · Getting started

Easy · 24 questions

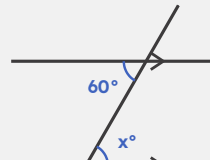
WE DO With your teacher: copy what they write on the board.

Example 1 Mark the alternate angle, then find the pronumeral.

a Mark the angle alternate to the one shown.



b Find x .



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

REMEMBER **ALTERNATE** angles are on opposite sides of the transversal.

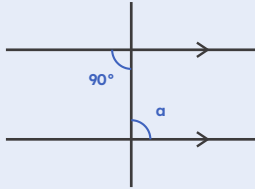
ROUND 1 Questions 1–12: the lines are parallel. Alternate angles are equal. Find the alternate angle.

- | | | | |
|-------------------------------|-------------------------------|--------------------------------|-------------------------------|
| 1 111°
_____ | 2 28°
_____ | 3 78°
_____ | 4 119°
_____ |
| 5 91°
_____ | 6 75°
_____ | 7 105°
_____ | 8 139°
_____ |
| 9 113°
_____ | 10 42°
_____ | 11 133°
_____ | 12 68°
_____ |

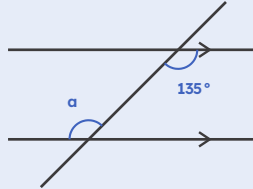
YOU DO On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

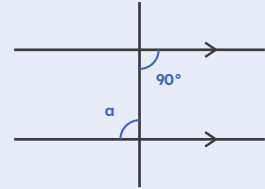
ROUND 2 Questions 13–24: the lines are parallel. Find a .



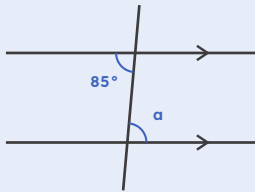
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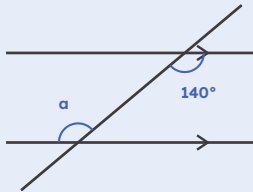
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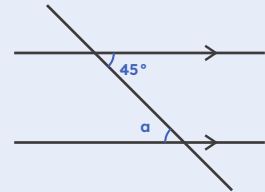
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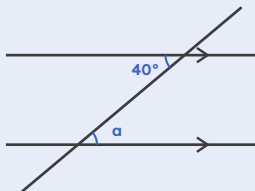
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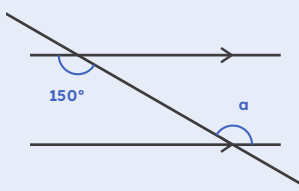
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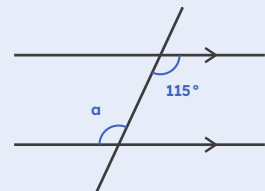
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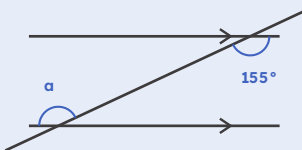
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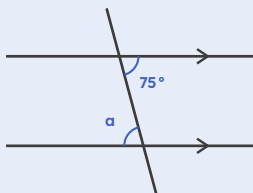
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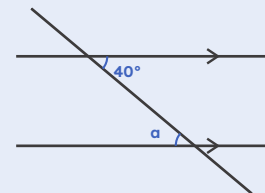
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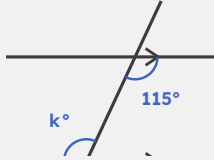
Level 2 · Building up

Medium · 31 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find k , then answer the question.

a Find k .



Your teacher works this out. Copy every step.

b Alternate angles are between the parallel lines. True or false?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

31 QUESTIONS

REMEMBER They are between the parallel lines (Z shape).

ROUND 1 Questions 1–15: the lines are parallel. Find the alternate angle (a) and the co-interior angle (c).

1 83°

2 13°

3 117°

4 43°

5 109°

6 52°

7 62°

8 151°

9 70°

10 37°

11 30°

12 23°

13 51°

14 50°

15 39°

YOU DO On your own: **Medium** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2 Questions 16–31: alternate angles are equal. Find x .

16 $2x + 21 = 87$

17 $4x + 13 = 69$

18 $2x + 25 = 81$

19 $3x + 8 = 35$

20 $2x + 17 = 71$

21 $4x + 1 = 109$

22 $3x + 23 = 59$

23 $3x + 7 = 121$

24 $4x + 20 = 96$

25 $3x + 4 = 94$

26 $4x + 1 = 97$

27 $4x + 26 = 134$

28 $2x = 78$

29 $4x + 9 = 45$

30 $2x + 9 = 45$

31 $2x + 20 = 94$

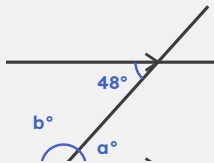
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find a and b , then explain.

a Find a and b .



Your teacher works this out. Copy every step.

b Why is it called a Z shape?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER On parallel lines they are **EQUAL**.

1 Draw a Z shape and mark a pair of alternate angles.

2 Alternate angles are $(x + 30)^\circ$ and 75° . Find x .

3 In Set C question 3, what two rules did you use?

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Alternate angles are $5x^\circ$ and $(2x + 45)^\circ$. Find x and the angle size.

1. Alternate angles are equal
2. $5x = 2x + 45$, so $3x = 45$
3. $x = 15$, angle = 5×15

Answer: $x = 15$, angle 75°

YOU DO

On your own: read the worked example, then try these.

- E1** Alternate angles are $(4a - 20)^\circ$ and $(2a + 30)^\circ$. Find a .

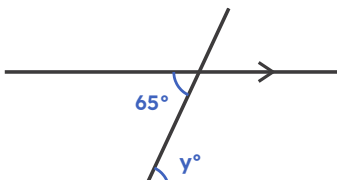
- E2** Alternate angles are $3m^\circ$ and $(m + 70)^\circ$. Find both angles.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

Find y .

Level 2

What letter shape do alternate angles make?

Level 3

Alternate angles are $(3x - 15)^\circ$ and 90° . Find x .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Alternate angles are on opposite sides of the transversal.
 2. They are between the parallel lines (Z shape).
 3. On parallel lines they are **equal**.
 4. Reason: “alternate angles, $AB \parallel CD$ ”.

Level 1

Alternate to top lower-left?

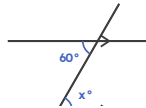
1. Other side, other line, inside

Answer: bottom upper-right

Level 2

Find x

1. Alternate angles



Answer: $x = 60$

Level 3

Find a and b

1. $a = 48$ alternate
2. $b = 180 - 48$

Answer: $a = 48$, $b = 132$

Extension

$$5x = 2x + 45$$

1. $3x = 45$

Answer: $x = 15$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.09

Co-interior angles

Date _____

Learning intentions

We are learning to:

- identify co-interior angles on parallel lines

Success criteria

I can:

- ☐ find **co-interior** angles (C shape)
- ☐ know they add to 180° on parallel lines
- ☐ find unknown angles using them

Key terms Write the definition as your teacher explains it.

Co-interior angles

Supplementary

Notes

Level 1 · Getting started

Easy · 24 questions

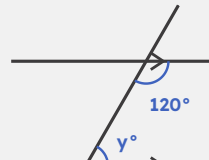
WE DO With your teacher: copy what they write on the board.

Example 1 Mark the co-interior angle, then find the pronumeral.

a Mark the angle co-interior to the one shown.



b Find y .



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

REMEMBER **CO-INTERIOR** angles are on the same side of the transversal.

ROUND 1 Questions 1–12: the lines are parallel. Co-interior angles add to 180° . Find the co-interior angle.

1 149°

2 133°

3 101°

4 72°

5 106°

6 76°

7 68°

8 79°

9 89°

10 23°

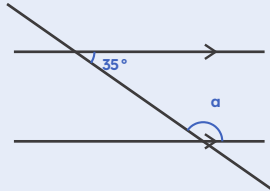
11 80°

12 43°

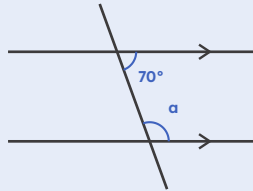
YOU DO On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

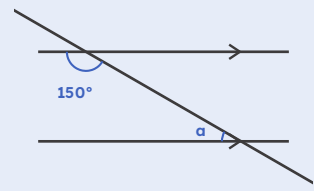
ROUND 2 Questions 13–24: the lines are parallel. Find a .



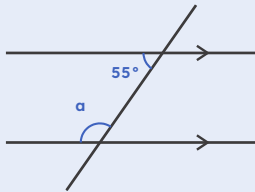
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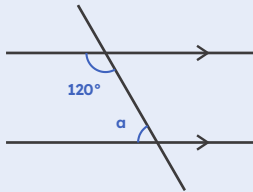
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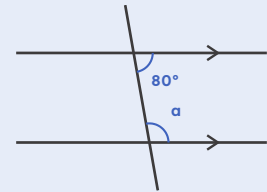
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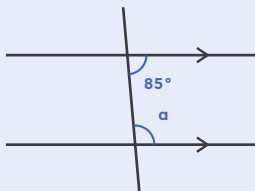
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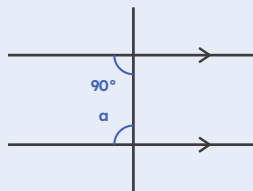
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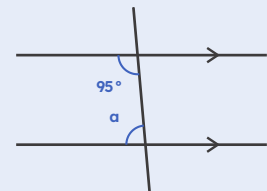
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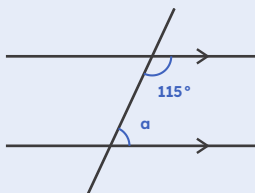
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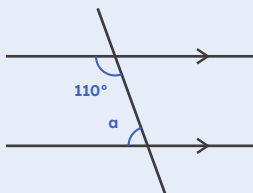
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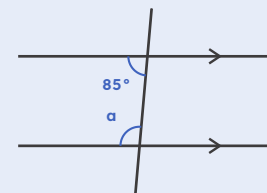
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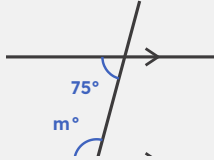
Level 2 · Building up

Medium · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find m , then answer the question.

a Find m .



Your teacher works this out. Copy every step.

b Are co-interior angles equal? Explain.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER They are between the parallel lines (C shape).

ROUND 1 Questions 1–16: co-interior angles on parallel lines. Find y .

1 $y^\circ + y^\circ + 84^\circ = 180^\circ$

2 $y^\circ + y^\circ + 148^\circ = 180^\circ$

3 $y^\circ + y^\circ + 116^\circ = 180^\circ$

4 $y^\circ + y^\circ + 138^\circ = 180^\circ$

5 $y^\circ + y^\circ + 80^\circ = 180^\circ$

6 $y^\circ + y^\circ + 64^\circ = 180^\circ$

7 $y^\circ + y^\circ + 150^\circ = 180^\circ$

8 $y^\circ + y^\circ + 70^\circ = 180^\circ$

9 $y^\circ + y^\circ + 82^\circ = 180^\circ$

10 $y^\circ + y^\circ + 72^\circ = 180^\circ$

11 $y^\circ + y^\circ + 100^\circ = 180^\circ$

12 $y^\circ + y^\circ + 94^\circ = 180^\circ$

13 $y^\circ + y^\circ + 164^\circ = 180^\circ$

14 $y^\circ + y^\circ + 78^\circ = 180^\circ$

15 $y^\circ + y^\circ + 170^\circ = 180^\circ$

16 $y^\circ + y^\circ + 134^\circ = 180^\circ$

YOU DO On your own: **Medium** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2 Questions 17–32: are the lines parallel? The co-interior angles are given. Write yes or no.

17 130° and 50°

18 115° and 62°

19 140° and 40°

20 146° and 31°

21 129° and 43°

22 32° and 156°

23 37° and 149°

24 126° and 54°

25 71° and 109°

26 63° and 119°

27 80° and 96°

28 63° and 113°

29 120° and 57°

30 113° and 75°

31 86° and 94°

32 145° and 35°

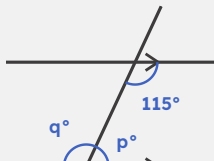
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find p and q , then name the three rules.

a Find p and q .



Your teacher works this out. Copy every step.

b Name the three angle pairs on parallel lines and their rules.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER On parallel lines they add to 180° .

1 Co-interior angles are x° and $2x^\circ$. Find x .

2 Co-interior angles are equal. What size are they? What does this tell you about the transversal?

3 Draw a C shape and mark co-interior angles.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Co-interior angles are $(2x + 10)^\circ$ and $(3x - 30)^\circ$. Find x .

1. Co-interior angles add to 180°
2. $2x + 10 + 3x - 30 = 180$
3. $5x - 20 = 180$, $5x = 200$

Answer: $x = 40$

YOU DO

On your own: read the worked example, then try these.

- E1 Co-interior angles are $4y^\circ$ and $5y^\circ$. Find both angles.

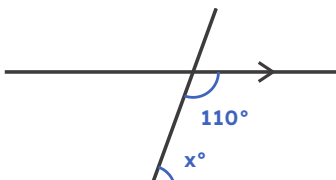
- E2 Co-interior angles are $(a + 25)^\circ$ and $(2a + 5)^\circ$. Find a .

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____



Level 1

Find x .

Level 2

What do co-interior angles add to on parallel lines?

Level 3

Co-interior angles are $3y^\circ$ and $6y^\circ$. Find y .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Co-interior angles are on the same side of the transversal.
 2. They are between the parallel lines (C shape).
 3. On parallel lines they add to 180° .
 4. Reason: "co-interior angles, $AB \parallel CD$ ".

Level 1

Co-interior to top lower-right?

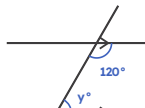
1. Same side, inside

Answer: bottom upper-right

Level 2

Find y

1. $y + 120 = 180$



Answer: $y = 60$

Level 3

p and q

1. $p = 180 - 115 = 65$
2. $q = 180 - 65$

Answer: $p = 65, q = 115$

Extension

$$(2x + 10) + (3x - 30) = 180$$

1. $5x = 200$

Answer: $x = 40$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.10

Finding unknown angles on parallel lines

Date _____

Learning intentions

We are learning to:

- use angle rules to find unknown angles on parallel lines

Success criteria

I can:

- ☐ choose the right parallel-line rule
- ☐ combine it with other angle rules
- ☐ give a reason for every step

Key terms Write the definition as your teacher explains it.

Reason

Parallel lines

Notes

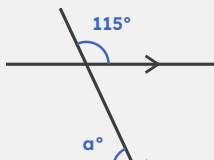
Level 1 · Getting started

Easy · 15 questions

WE DO With your teacher: copy what they write on the board.

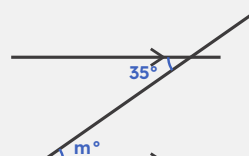
Example 1 Find the pronumeral. Give a reason.

a Find a . Give a reason.



Your teacher works this out. Copy every step.

b Find m . Give a reason.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

15 QUESTIONS

REMEMBER Corresponding and alternate angles are **EQUAL**.

ROUND 1 Questions 1–6: which rule? Write corresponding, alternate or co-interior.

1 C (or U) shape

2 the angles add to 180°

3 the angles are equal (Z)

4 same position at each crossing

5 a C shape facing left

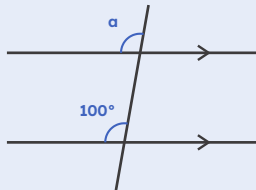
6 Z shape

YOU DO On your own: **Easy** keep going: Round 2.

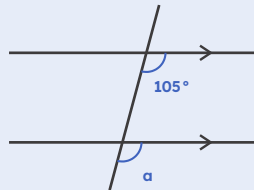
9 QUESTIONS MORE

ROUND 2 Questions 7–15: the lines are parallel. Find α .

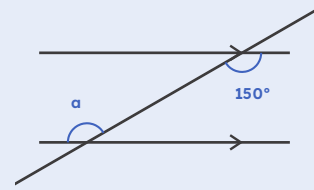
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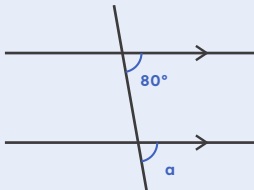
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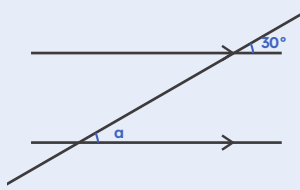
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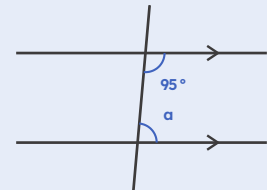
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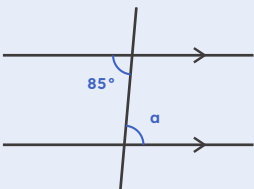
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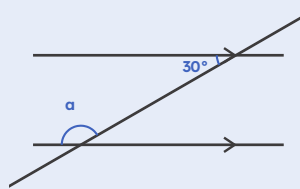
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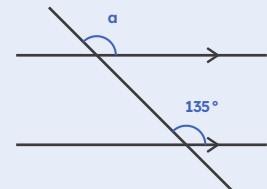
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14



15



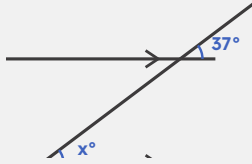
Level 2 · Building up

Medium · 15 questions

WE DO With your teacher: copy what they write on the board.

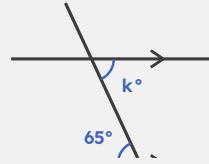
Example 2 Find each pronumeral. Give reasons.

a Find x and y .



Your teacher works this out. Copy every step.

b Find k .



Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

15 QUESTIONS

REMEMBER Co-interior angles add to 180° .

ROUND 1 Questions 1–12: the lines are parallel. One angle is given. Find the other three angles at the same crossing.

1 129°

2 52°

3 55°

4 31°

5 79°

6 104°

7 142°

8 18°

9 130°

10 16°

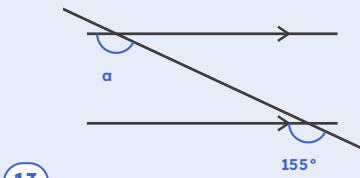
11 127°

12 141°

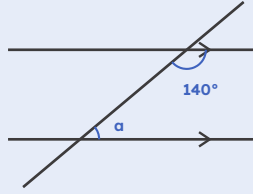
YOU DO On your own: **Medium** keep going: Round 2.

3 QUESTIONS MORE

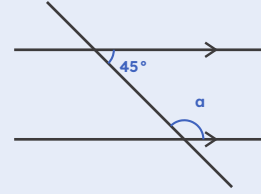
ROUND 2 Questions 13–15: the lines are parallel. Find a and give the rule.



13



14



15

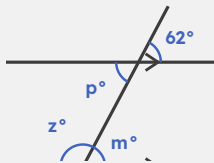
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find each pronumeral, then explain.

a Find p , m and z .



Your teacher works this out. Copy every step.

b Explain how you would find an angle if no rule gives it directly.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Harder diagrams.

3 QUESTIONS

REMEMBER Use straight lines (180°) and vertically opposite angles too.

1 Two parallel lines are cut by a transversal. One angle is 57° . Find all 8 angles.

2 An angle on parallel lines is 3 times its co-interior angle. Find both angles.

3 Explain why the 4 angles at the top intersection match the 4 at the bottom.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Find x if one angle is $(5x - 10)^\circ$ and its co-interior angle is $(3x + 30)^\circ$.

1. Co-interior angles add to 180°
2. $8x + 20 = 180$
3. $8x = 160$

Answer: $x = 20$ (angles 90° and 90°)

YOU DO

On your own: read the worked example, then try these.

- E1** Alternate angles are $(7x - 12)^\circ$ and $(4x + 36)^\circ$. Find x .

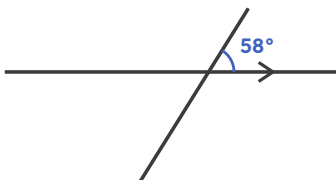
- E2** Corresponding angles are $(x + 15)^\circ$ and $(2x - 25)^\circ$. Find x .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

Find x .

Level 2

Give the reason you used.

Level 3

Find all 8 angles when one angle is 58° .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Corresponding and alternate angles are **equal**.
 2. Co-interior angles add to **180°** .
 3. Use straight lines (180°) and vertically opposite angles too.
 4. Give a **reason** for each step.

Level 1

Find a

1. Top ur $115^\circ \rightarrow$ bottom ur 115° (corr.)
2. $a = 180 - 115$ (straight line)

Answer: $a = 65$

Level 2

Find m

1. Alternate angles

Answer: $m = 35$

Level 3

p, m, z

1. $p = 62$ (vert. opp.)
2. $m = 62$ (corr.)
3. $z = 118$ (straight line)

Answer: $p = 62, m = 62, z = 118$

Extension

$$5x - 10 + 3x + 30 = 180$$

1. $8x = 160$

Answer: $x = 20$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.11

Proving lines are parallel

Date _____

Learning intentions

We are learning to:

- use angle rules to prove that two lines are parallel

Success criteria

I can:

- ☐ check whether corresponding or alternate angles are equal
- ☐ check whether co-interior angles add to 180°
- ☐ write a conclusion: $AB \parallel CD$ with a reason

Key terms Write the definition as your teacher explains it.

Prove

Converse

Notes

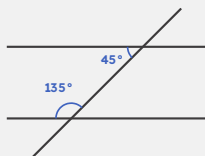
Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.

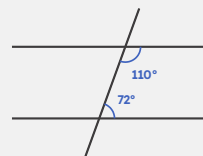
Example 1 Is $AB \parallel CD$? Give a reason.

a Is $AB \parallel CD$? Give a reason.



Your teacher works this out. Copy every step.

b Is $AB \parallel CD$? Give a reason.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

REMEMBER Corresponding angles are **EQUAL**, or

ROUND 1 Questions 1–9: is $AB \parallel CD$? Write yes or no.

1 Corresponding: 56° and 56°

2 Alternate: 93° and 93°

3 Co-interior: 139° and 41°

4 Co-interior: 145° and 35°

5 Co-interior: 73° and 73°

6 Corresponding: 36° and 36°

7 Corresponding: 33° and 33°

8 Corresponding: 79° and 101°

9 Alternate: 71° and 109°

YOU DO

On your own: **Easy** keep going: Round 2.

15 QUESTIONS MORE

ROUND 2

Questions 10–24: is $AB \parallel CD$? Write yes or no.10 Corresponding: 41° and 50°

11 Corresponding: 95° and 95°

12 Alternate: 71° and 79°

13 Alternate: 145° and 145°

14 Co-interior: 30° and 30°

15 Alternate: 134° and 46°

16 Co-interior: 113° and 67°

17 Co-interior: 126° and 54°

18 Alternate: 110° and 70°

19 Co-interior: 84° and 96°

20 Co-interior: 100° and 80°

21 Corresponding: 123° and 57°

22 Co-interior: 51° and 51°

23 Corresponding: 106° and 106°

24 Alternate: 119° and 119°

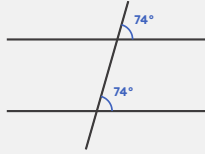
Level 2 · Building up

Medium · 23 questions

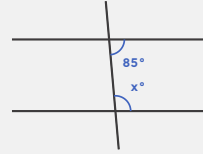
WE DO With your teacher: copy what they write on the board.

Example 2 Decide whether $AB \parallel CD$, then find the x that makes $AB \parallel CD$.

a Is $AB \parallel CD$? Give a reason.



b Which value of x proves $AB \parallel CD$?



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

23 QUESTIONS

REMEMBER alternate angles are **EQUAL**, or

ROUND 1 Questions 1–15: find x so that $AB \parallel CD$.

1 corresponding angles: 150° and x°

2 alternate angles: 45° and x°

3 alternate angles: 80° and x°

4 co-interior angles: 55° and x°

5 corresponding angles: 85° and x°

6 alternate angles: 125° and x°

7 corresponding angles: 35° and x°

8 co-interior angles: 150° and x°

9 co-interior angles: 60° and x°

10 co-interior angles: 115° and x°

11 alternate angles: 35° and x°

12 alternate angles: 150° and x°

13 corresponding angles: 40° and x°

14 corresponding angles: 125° and x°

15 co-interior angles: 120° and x°



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

A line crosses AB and CD. The vertically opposite angle to an angle on AB is 72° , and the alternate angle on CD is 72° . Is $AB \parallel CD$?

1. Vertically opposite angles are equal, so the angle on AB is 72°
2. Alternate angles 72° and 72° are equal

Answer: Yes, $AB \parallel CD$ (alternate angles equal)

YOU DO

On your own: read the worked example, then try these.

- E1 Angles on a straight line on AB are 48° and x° . The co-interior angle to x° on CD is 48° . Is $AB \parallel CD$?

- E2 Why is it not enough to measure the angles with a protractor to prove lines are parallel?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Alternate angles are 62° and 62° . Are the lines parallel?

Level 2

Co-interior angles are 105° and 85° . Are the lines parallel? Explain.

Level 3

Find x so that corresponding angles $(2x + 6)^\circ$ and 80° make $AB \parallel CD$.

Lesson summary

Missed the lesson? Start here.

Tests for parallel lines

1. Corresponding angles are **equal**, or
3. co-interior angles add to **180°**.

2. alternate angles are **equal**, or
4. Write: “AB $\parallel$ CD (alternate angles are equal)”.

Level 1

135° and 45° co-interior?

1. $135 + 45 = 180$

Answer: AB $\parallel$ CD

Level 2

110° and 72° co-interior?

1. $110 + 72 = 182 \neq 180$

Answer: not parallel

Level 3

74° and 74° corresponding

1. Equal

Answer: AB $\parallel$ CD

Extension

Co-interior 85° and x°

1. $x = 180 - 85$

Answer: $x = 95$

My reflection: what did I learn today? Which question was hardest, and why?

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